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RESEARCH ARTICLE

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Has Strategic Purchasing Led to Improvements in Health Systems? A Narrative Review of Literature on Strategic Purchasing

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ABSTRACT

Strategic purchasing is noted in the literature as an approach that can improve the efficiency of health spending, increase equity in access to health care services, improve the quality of health care delivery, and advance progress toward universal health coverage. However, the evidence on how strategic purchasing can achieve these improvements is sparse. This narrative review sought to address this evidence gap and provide decision makers with lessons and policy recommendations. The authors conducted a systematic review based on two research questions: 1) What is the evidence on how purchasing functions affect purchasers' leverage to improve: resource allocation, incentives, and accountability; intermediate results (allocative and technical efficiency); and health system outcomes (improvements in equity, access, quality, and financial protection)? and 2) What conditions are needed for a country to make progress on strategic purchasing and achieve health system outcomes? We used database searches to identify published literature relevant to these research questions, and we coded the themes that emerged, in line with the purchasing functions—benefits specification, contracting arrangements, provider payment, and performance monitoring—and the outcomes of interest. The extent to which strategic purchasing affects the outcomes of interest in different settings is partly influenced by how the purchasing functions are designed and implemented, the enabling environment (both economic and political), and the level of development of the country's health system and infrastructure. For strategic purchasing to provide more value, sufficient public funding and pooling to reduce fragmentation of schemes is important.

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KEYWORDS

Africa; provider payment; purchasing; strategic health purchasing; Strategic Health Purchasing Progress Tracking Framework

Introduction

Strategic health purchasing is noted in the literature as an approach that can improve the efficiency of health spending, increase equity in access to health care, improve the quality of health care delivery, and advance progress toward universal health coverage (UHC).^{1–6} Strategic purchasing involves a systematic evaluation of health needs, benefit package design, selection of appropriate providers, payment incentives to providers, and provider monitoring to encourage good performance, using available pooled funds.⁷

The Strategic Health Purchasing Progress Tracking Framework developed by the Strategic Purchasing Africa Resource Center (SPARC) and its technical partners describes the purchasing functions, capacities, and enabling environment required for strategic purchasing to improve health outcomes (Figure 1).⁸ The premise underlying the framework is that effective strategic purchasing involves a set of core functions (benefits specification, contracting arrangements, provider payment, and performance monitoring) that are supported by

clear institutional arrangements that allocate responsibility for carrying out those functions, governance structures that provide oversight and accountability, and mechanisms to ensure effective stakeholder participation. The core functions are also influenced by external factors that can enhance or limit purchasing power, including the legal and regulatory environment governing other aspects of the health system, the share of the population covered by the purchaser and the share of total health spending it manages, public financial management (PFM) rules, and the market structure of purchasers and providers. Purchasing capacities further affect how well each function is carried out and thus how much they can influence purchasers' leverage to improve resource allocation, incentives, and accountability; and achieve the desired intermediate results and health system outcomes.

Application of the framework in nine sub-Saharan African countries revealed that progress in strategic purchasing has been limited to certain core functions in some schemes, and that progress at the scheme level

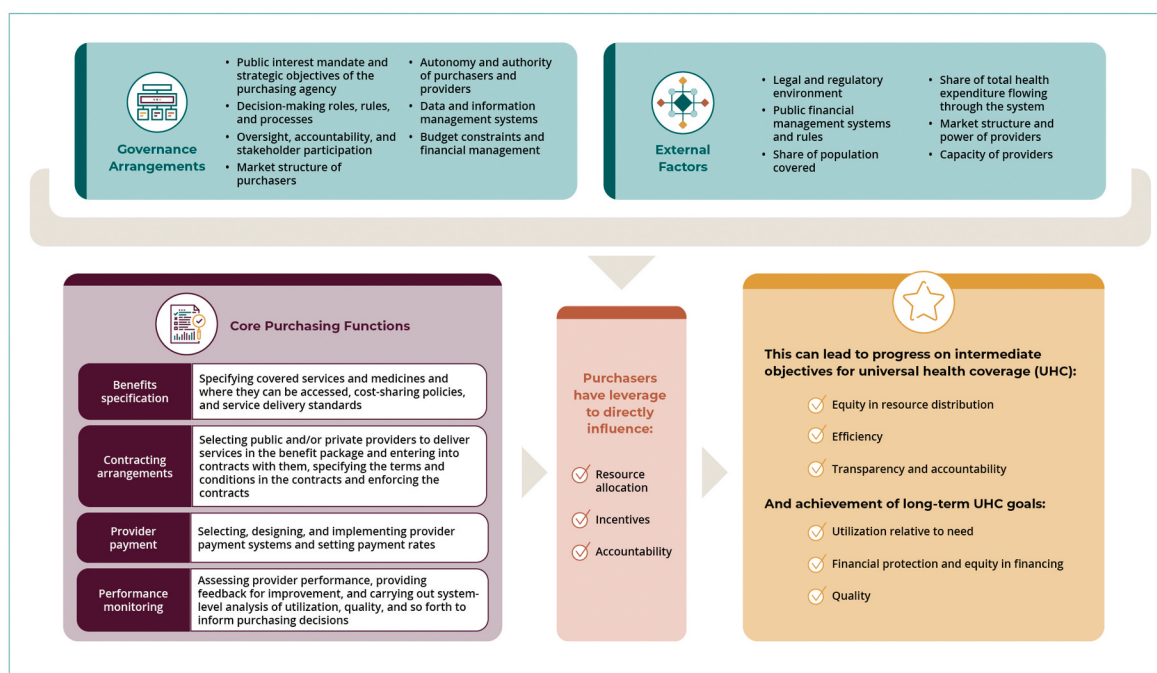


Figure 1. Strategic Health Purchasing Progress Tracking Framework.

has not resulted in large-scale health system change.⁹ One reason is fragmentation of health financing arrangements, which results in purchasers having little leverage to improve resource allocation and provide incentives to health providers to improve their performance and provide good-quality care. Despite a growing body of evidence on strategic purchasing, no systematic study has been conducted to learn how strategic purchasing reforms have worked in low- and middle-income countries, including key factors that have enabled success, how those factors have affected purchasing functions, and how those functions have led to results. This paper seeks to address that evidence gap.

In this paper, we examine the evidence on country experience in strengthening the four core purchasing functions identified in the framework—benefits specification, contracting arrangements, provider payment, and performance monitoring—and whether and how improving these core functions has led to direct effects on resource allocation, incentives, accountability, and ultimately intermediate results and health system outcomes. We also examine the preconditions that have led to or contributed to the observed effects and key lessons learned.

Methods

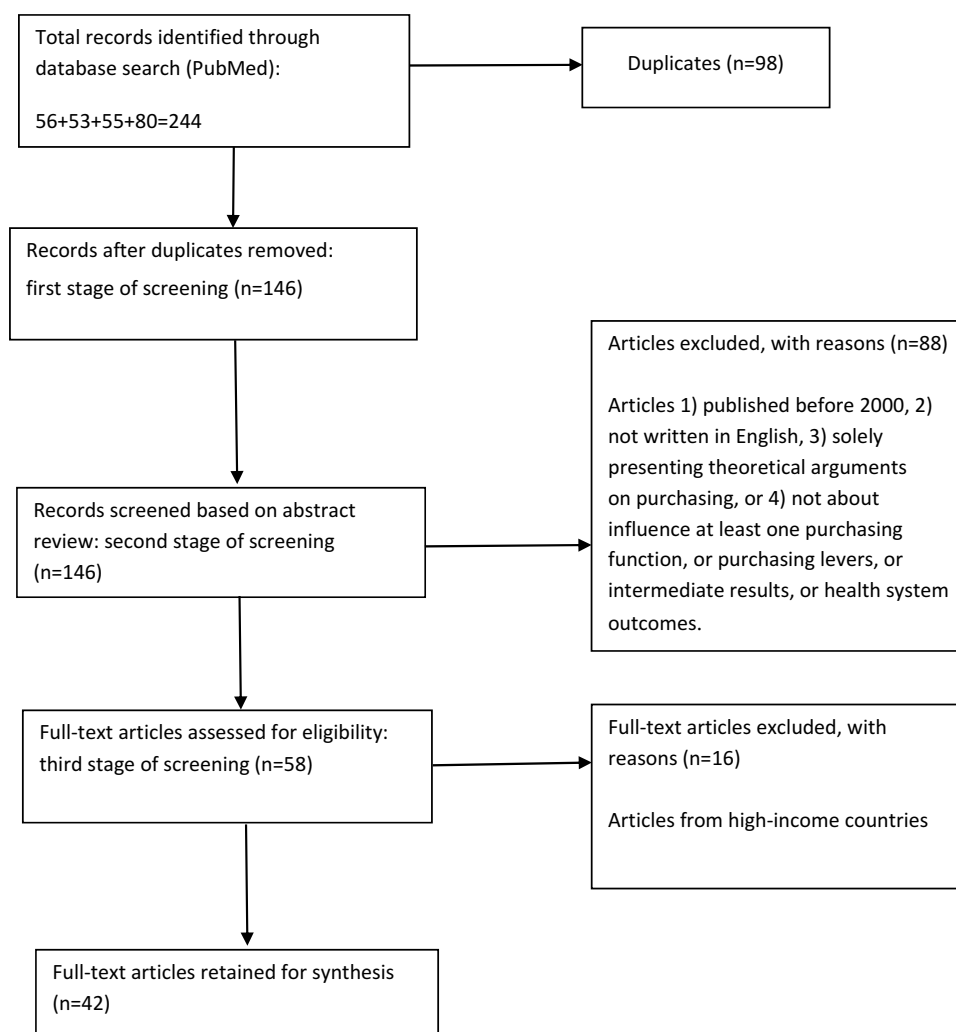
We conducted a literature review between May and August 2020 based on two research questions:

- What is the evidence on how the core purchasing functions (benefits specification, contracting arrangements, provider payment, and performance monitoring) affect purchasers' leverage to improve: resource allocation, incentives, and accountability; intermediate results (allocative and technical efficiency); and health system outcomes (improvements in equity, access, quality, and financial protection)?^{8,10,11}
- What conditions are needed for a country to make progress on strategic purchasing and achieve the desired intermediate results and health system outcomes?

We conducted database searches to identify published literature relevant to these research questions. (See [Table 1](#).) We exported the outputs from the searches to a citation manager (Zotero) and uploaded them to EPPI-Reviewer Web (version 4.11.4.0). After screening for duplicates, one reviewer completed an independent review of the title and abstracts. The exclusion criteria were: articles published before 2000; articles not written in English; articles that did not influence at least one purchasing function and describe effects on purchasers leverage to improve resource allocation, incentives, and accountability, intermediate results, or health system outcomes; articles that solely presented theoretical arguments on the topics of interest (e.g., articles on the economic basis of purchasing). The

Table 1. Terms and boolean operators used in the literature search.

Strategic Purchasing Function	Search Terms	# of PubMed Results
Benefits specification	(health benefit* OR package OR entitlement*) AND (access OR outcome* OR utilization OR quality OR efficient* OR equity* OR financial protection) AND strategic purchasing AND health	56
Contracting arrangements	(contract*) AND (access OR outcome* OR utilization OR quality OR efficient* OR equity* OR financial protection) AND strategic purchasing AND health	53
Provider payment	(payment OR results-based OR output-based OR performance-based OR PBF OR RBF OR payment mechanism OR pay-for-performance OR P4P) AND (access OR outcome* OR utilization OR quality OR efficient* OR equity* OR financial protection) AND strategic purchasing AND health	55
Performance monitoring	(monitor* OR autonomy OR accountability OR governance) AND (access OR outcome* OR utilization OR quality OR efficient* OR equity* OR financial protection) AND strategic purchasing AND health	80
Total		244

**Figure 2.** PRISMA diagram showing the selection of articles for review.

final list for review included 42 articles from Africa, Asia, and Latin America that covered at least one of the four purchasing functions. (See [Appendix A](#) for the full list.) [Figure 2](#) depicts the study selection process.

The articles selected for review used different frameworks to describe the purchasing functions.

However, the core purchasing functions as defined in the Strategic Health Purchasing Progress Tracking Framework are based on the key purchasing decisions—what to buy (benefits specification), from whom to buy (contracting arrangements), and how to buy (provider payment), and using information to improve these decisions (performance monitoring).

The authors used these key principles to match the description in the papers to the core purchasing functions—benefits specification, contracting arrangements, provider payment, and performance monitoring. While all 42 papers described at least one of the purchasing functions, 29 of the 42 papers described two or more of these functions ([Appendix A](#)). Of the 42 papers, only 5 had descriptions of all the core purchasing functions. The authors supplemented the 42 papers with additional country reports from the European Observatory on Health Systems and Policies, World Bank and World Health Organization.

Common themes were extracted by authors based on the purchasing function and associated results were summarized for each paper according to the purchasing function and outcome of interest ([Appendix B](#)). The results of interest were effects of purchasing functions on purchasers' leverage to improve: resource allocation, incentives, and accountability; intermediate results (allocative and technical efficiency referred to broadly as efficiency); and health system outcomes ([Figure 1](#)).

Results

The authors mapped the findings to the four purchasing functions and summarized how improving these functions affected purchasers' leverage and intermediate results, as well as health system outcomes. (See [Appendix B](#).)

This section summarizes the underlying conditions and enabling factors that contribute to the observed results and outcomes.

Question 1. What is the evidence on how the core purchasing functions affect purchasers' leverage, intermediate results, and health system outcomes?

Core Function #1: Benefits Specification

Benefits specification includes determining the services and interventions in the benefit package, the service delivery standards, where and how the services will be accessed (including gatekeeping policies), how much of the cost of services will be covered by the purchaser (and accompanying cost-sharing policies), and which medicines will be covered.⁸ Specifying the covered services and medicines and how beneficiaries can access them is the first opportunity for purchasers to improve resource allocation, incentives, and

accountability, by directing resources to the highest-priority services and populations.

Benefits specification can lead to improvements in equity and access to services when the benefit package is informed by evidence, is needs based, and takes into account the preferences and values of the covered population. The benefit packages in Thailand and Mexico were informed by an evidence-based health technology assessment,^{12–15} but processes in the Philippines and Vietnam have been less transparent and more subjective.^{14,16} These differences in benefit package design partly explain the variation in effects on health care access and other health system outcomes in those countries.

Comprehensive benefit packages such as those in the national health insurance systems of Ghana, Mexico, and Thailand—which include necessary inpatient, outpatient, health promotion, and disease prevention services for the enrolled population—have been instrumental in achieving health system outcomes in those countries, including increasing equitable access to and utilization of services and improving financial protection.^{4,8,17,18} Clearly defined and comprehensive benefit packages have enabled effective resource allocation and accountability for the schemes in meeting commitments to the population.¹⁹ Benefit packages can exacerbate inequity, however, when they vary significantly across enrolled population groups, as has been the case in Kenya and the Philippines.^{16,20}

Benefit packages should be adequately communicated to the population so people are aware of their entitlements and obligations and purchasers and providers can be held accountable for providing those entitlements.⁷ Effective communication empowers beneficiaries—particularly marginalized groups—to claim their benefits while increasing care utilization overall and reducing possible variations in access to care.²⁰ This is particularly true with respect to cost-sharing obligations on the part of beneficiaries. In the Kyrgyz Republic, for example, the introduction of a clearly defined and communicated State Guaranteed Benefit Package in 2001 included formal copayments for nonexempt population groups. Although this was the first time that formal cost-sharing was included in the government health system, this approach has led to documented improvements in financial protection, access, and efficiency and a reduction in informal payments.²¹

Finally, the benefit package should align with the capacity of the service delivery system to meet commitments to the population. A mismatch between the benefit package and provider capacity to deliver the services hampers access to services and can erode trust in the

government's commitment to ensure access to services with financial protection.^{8,21}

Core Function #2: Contracting Arrangements

Contracting arrangements include systems and policies for selecting public and/or private providers to deliver services in the benefit package, contracts that specify terms and conditions (e.g., at which level specific services can be delivered and data reporting requirements), and enforcement of the contracts.⁸ Contracting can directly contribute to improved resource allocation, incentives, and accountability by specifying which services can be delivered by which providers, the payment terms, and how providers will be held accountable for complying with the agreed-upon terms. Contracts are an important tool for communicating expectations and introducing a credible threat of being excluded from financing if a provider does not meet minimum quality and performance standards.²²

Contracting is most effective when clear criteria are specified for engaging providers and a clear process is in place for identifying providers and establishing and managing agreements with them.²³ When population needs are considered in designing the terms of contracts with providers, contracting can enable improvements in health system outcomes. In Thailand's Universal Coverage (UC) Scheme and China's social health insurance system, providers are contracted to provide a specific range of services to a specified population.^{4,12} A "close-to-the-client" provider is contracted to provide outpatient services to the catchment population, to reduce travel costs for patients and increase utilization.

Contracting with private providers is a strategy used by many government purchasing agencies to increase access to services and ensure quality of care.¹⁷ In Tanzania, for example, access to publicly funded services in remote areas has improved through contracting with faith-based providers, and financial protection has improved by reducing out-of-pocket payments to these providers.¹⁷

Core Function #3: Provider Payment

Provider payment includes the systems and policies for selecting, designing, and implementing provider payment systems and setting payment rates.⁸ Provider payment is the main tool that purchasers have to improve incentives for providers, but payment systems can also influence resource allocation, either directly or indirectly through incentives, and accountability by linking payment to service delivery.

The evidence shows that provider payment remains an underused strategic purchasing tool, with many

countries continuing to rely on input-based budgets or open-ended fee-for-service payment, neither of which has been found to lead directly to improved health system outcomes. Input-based budget payment typically lacks flexibility for providers to allocate spending between line items, with savings made in one line not easily moved to cover deficits in others.²⁴ On the other hand, open-ended fee-for-service payment can lead to cost escalation and a shift to more expensive services.^{12–15}

Provider payment can facilitate improved outcomes when payment is linked to services in the package ("output-based payment") and specific service delivery objectives. This creates incentives for efficient and high-quality service delivery, promotes effective allocation of resources across levels of care, and enables management of the purchaser's budget—that is, payments are not open ended but are capped at some level of the system.^{12,25}

Some countries are moving toward output-based payment methods, such as capitation and case-based payment using diagnosis-related groups (DRGs), with some evidence of positive results.^{12,26} In 2009, China changed the way six public hospitals were paid, moving from fee-for-service to blended methods that included case-based payment with DRGs. DRG payment led to reductions of 6.2% in government health expenditure and 10.5% in out-of-pocket payments by patients per hospital admission. No evidence was found of increased hospital readmission rates or cost shifting from cases eligible for DRG payment to ineligible cases.²⁷

Although stand-alone performance-based financing schemes show mixed results,²⁸ blending performance-based incentives with appropriately selected and designed output-based payment systems have brought service delivery improvements in Estonia.²⁹

Further, it is important to align the payment incentives from multiple funding flows and maintain trust between purchasers and providers—for example, through timely payment and routinely revised payment rates.^{30–32} A review of provider payment for primary care physicians in China showed that a mixed system of input-based and output-based payment to health facilities was more effective under a long-term contract with the government.³⁰ Regardless of the payment system, when providers are not paid at predefined times, they lose motivation and have an incentive to stop providing services or to charge informal fees.³¹

Core Function #4: Performance Monitoring

Performance monitoring includes systems and processes for assessing provider performance, providing feedback

for improvement, and carrying out system-level analysis of utilization and quality, to inform purchasing decisions.⁸ Strong performance monitoring systems have been shown to improve health provider accountability, leading to significant gains in efficiency, better adherence to treatment guidelines, expanded access to health care, and improved quality of care.^{8,12,33} Appropriately monitoring contract performance is important for service quality.¹⁷ Defining and tracking metrics for improvement can result in improvements in quality and performance.³³

Using performance monitoring to enable improvements in health system results requires strong data systems and effective processes to provide feedback to providers.⁷ For example, reporting requirements and data quality should be made explicit in contracts with providers.³⁴ Monitoring information should be shared with providers, along with supportive feedback, to enable dialog between purchasers and providers and support performance improvement.³⁵ Further, the intensity of monitoring and verification should be balanced with what these measures can help accomplish in terms of improved accountability and provider performance.³⁶ Monitoring and verification data should be used for further system-level analysis to monitor trends, whether objectives are being met, and whether purchasing policies are leading to any unintended consequences.

Community engagement in the design of performance monitoring systems improves provider accountability and adherence to contracting terms, and it ultimately increases access, equity, and patient satisfaction.³³ On the other hand, weak financial and human capacity can limit the effectiveness of performance monitoring.^{11,17,37}

Question 2. What conditions are needed for a country to make progress on strategic purchasing and achieve the desired intermediate results and health system outcomes?

A number of factors external to purchasing arrangements can affect the ability of strategic purchasing to facilitate improvements in intermediate results and health system outcomes. These factors include PFM systems and their rules for planning and budgeting, budget execution, and accounting for public funds.⁸ In many settings, PFM rules limit the use of output-based payment, such as restrictions on accounting for inputs rather than outputs and prepayment for goods and services. In Kenya and Argentina, facility improvements (including upgrades to infrastructure and equipment) and overall strengthening of service delivery and quality of care were seen when PHC providers received flexible

funds.^{12,26} However, these gains were reversed in Kenya when facility autonomy was reversed and budget execution decisions were recentralized to the county level.²⁷

Evidence from Uganda and Tanzania shows that a good budget structure with PFM rules that allow provider autonomy and flexibility can facilitate strategic purchasing and create a system of accountability for achieving health system goals. The bottom-up budgeting and planning process in Tanzania, which starts at the facility level, and program-based budgeting in Uganda help allocate funds based on priorities and population needs.³⁸ A good government budgeting process with PFM rules that allow for use of output-based payment can improve provider autonomy to use funds flexibly, while ensuring high levels of accountability, as well as improve allocation of resources for health to achieve improved efficiency and the health system goals of improved access and equity.

Purchasing functions, in concert with an enabling environment, can improve purchasers' leverage, intermediate results, and health system results, as seen in Argentina and Thailand.²⁶ In Argentina, only 0.5% of provincial expenditure for performance contracting by provinces led to reductions in newborn and child mortality. Providing incentives to the provincial health level and providers, empowering providers to control the use of new funds, and creating a culture of accountability fostered the program's success.

Comprehensive Strategic Purchasing: The Case of Argentina's Programa Sumar

Argentina's Plan Nacer, now Programa Sumar, is one example in which rigorous evidence demonstrates the effects of strategic purchasing implemented over time. The combination of changes to the four core purchasing functions has improved purchasers leverage, intermediate results and health system outcomes.

Benefits Specification

Programa Sumar provides a comprehensive benefit package, which has expanded incrementally over time and now covers neonatal care, immunizations, treatment of childhood conditions, reproductive care, and other essential services over the life cycle, from childhood through adolescence, reproductive age, adulthood, and old age.^{39,40}

Contracting Arrangements

Provider contracts define the services, payment arrangements, and performance metrics to be monitored.

Providers are granted autonomy in financial decision making, giving them the flexibility to respond to incentives.

Provider Payment

Resource allocation to the provinces is based on health indicators and provincial performance, while payment to facilities is made through fee-for-service payments based on achievement of predefined health indicators and targets.

Performance Monitoring

An impact evaluation of Plan Nacer showed that in participating facilities between 2005 and 2008, the risk of neonatal death among covered babies declined by 74%.^{39,41} The risk of low birth weight in participating facilities declined by 9% among non-beneficiaries and 19% among beneficiaries. Women enrolled in the plan were 20% less likely to need a cesarean section, and Plan Nacer averted an estimated 773 neonatal deaths and 1,071 cases of low-birth-weight babies and saved 25,401 total disability-adjusted life years (DALYs), with an estimated cost per lost DALY averted of USD \$814, well under Argentina's gross domestic product per capita of USD \$6,075.

Discussion

This review found cases of improvements in access to services, quality of care, and equity following implementation of one or a combination of the core purchasing functions. However, a key limitation is that the studies did not follow a randomized controlled trial (RCT) methodology that would be able to attribute specific improvements to strategic purchasing or attribute specific improvements to purchasing functions. This review was also limited because the authors only included papers written in English. Majority of the papers reviewed were based on health insurance schemes and a limited number of studies assessed purchasing through government-budgets. The findings are based largely on anecdotal evidence as synthesized from the studies, and the conclusions drawn in this review should be understood from that perspective. Further, the reviewed studies did not use similar frameworks to define the purchasing functions, such as the Strategic Health Purchasing Progress Tracking Framework. Many of the studies were descriptive in nature, describing one or more functions but not exactly in the same way as in the Strategic Health Purchasing Progress Tracking Framework. It is impossible to precisely attribute health system effects to strategic purchasing, and many of the examples in the

literature are anecdotal or inferred based on expected pathways for strategic purchasing to achieve the health system results of equitable access to good-quality services without financial hardship.

This review found few exemplars of well-functioning strategic purchasing systems among low- and middle-income countries. Differences in institutional capacities and health system infrastructure are among the key contextual factors that explain the varying levels of success with different interventions.

Nonetheless, there is evidence that even incremental progress can lead to improved health system outcomes and that, conversely, weak purchasing arrangements can work against UHC objectives even as coverage expands and health expenditure increases.⁸ An enabling environment is critical to achieving the benefits of strategic purchasing, particularly a strong PFM and budgeting system and autonomy of health providers to respond to purchasing incentives.

Furthermore, for strategic purchasing to work well, other health financing functions, including revenue collection and pooling, must be structured in a way that maximizes the benefits.^{10,11,42} Low public financing remains the main barrier to achieving UHC goals.⁴³ The health system must also have sufficient health workers, commodities, and medical supplies as well as strong leadership that makes evidence-based decisions. Fragmentation, which is sometimes created by donor-driven agendas, is a critical roadblock to sustainable systemwide strategic purchasing reforms.⁹ Even when progress is made in strategic purchasing within a scheme, only a small portion of the funds are channeled through the scheme, which limits the use of purchasing power to exert influence on the system.

This review has exposed key gaps in the evidence on the role of strategic purchasing in advancing progress toward better health system outcomes. More evidence is needed on how a combination of reforms to the purchasing functions can affect health system performance and, in particular, how the purchasing functions reinforce one another. Another area that needs more understanding is how to achieve the right power balance between purchasers and providers so they can engage effectively in purchasing functions and respond to incentives. There is also limited evidence on effective frameworks for engaging the private sector and improving contracting with private providers, especially in countries that largely finance health services through public resources.

Conclusion

Strategic purchasing has been identified as a key approach for achieving the UHC goals of improved

and equitable access and financial risk protection. However, improvements in purchasing depend on a range of factors, including strengths and shortcomings in human resources, information technology, governance, supply chains, and the regulatory environment. They also depend on external factors such as monitoring and accountability frameworks and the managerial autonomy of providers and purchasers. Although the evidence base is not broad, a number of lessons have emerged to guide countries that are looking to design or modify purchasing functions in a manner that can promote their health system goals.

Clear health system objectives and a strategy that clarifies the role of purchasers and the intended results are critical. Consensus among stakeholders on these objectives, including tradeoffs in benefits specification, contracting arrangements, and provider payment, is a prerequisite to aligning stakeholders around health system objectives. Social accountability and social justice mechanisms that promote transparency and citizen engagement are needed to improve the accountability of purchasers to beneficiaries. Effective communication is vital to inform the population of their entitlements and obligations; empower beneficiaries, particularly marginalized groups, to claim their benefits; improve utilization among marginalized groups; and reduce variations in access to care. Purchasers also need the capacity to carry out strategic planning and policy development, generate information on health needs and system capacity, make purchasing decisions within the constraints of the current system, and communicate strategically with both providers and the covered population. The involvement of high-level public health officials in designing strategic purchasing interventions, as well as a strong political and technical teams, can increase the chances of success of improving purchasing functions.

Although the evidence base remains incomplete, strategic purchasing is a necessary policy direction for countries at all income levels to channel limited health funds most effectively toward improved health system outcomes. Stronger evidence is needed on what makes strategic purchasing effective in different contexts to continue making the case for strategic purchasing and to better understand how improvements in purchasing can result in better health system outcomes.

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Author Contributions

FM and RS carried out the data extraction and synthesis. FM and A G-M wrote the first draft of the manuscript.

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Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article and/or its supplementary materials.

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Appendix A. Articles Included in the Full Text Review

Item	Title	Presence in sub-searches (1 = yes, blank = no)				
		Original search area result	Benefits specification	Contracting arrangements	Provider payment	Performance monitoring
1	Bastani (2015) (ID:48336806)	Resource allocation and purchasing arrangements to improve accessibility of medicines: Evidence from Iran.			1	
2	Bastani (2016) (ID:48336809)	Components that affect the implementation of health services' strategic purchasing: A comprehensive review of the literature.			1	
3	Behzadifar (2020) (ID:48325425)	The barriers to the full implementation of strategic purchasing and the role of health policy and decision-makers: Past, current status, ethical aspects and future challenges.	1	1		
4	Bigdeli (2009) (ID:48336813)	Barriers to access and the purchasing function of health equity funds: Lessons from Cambodia.			1	
5	Criel (2020) (ID:48325348)	Can mutual health organizations influence the quality and the affordability of healthcare provision? The case of the Democratic Republic of Congo.				1
6	Duran (2019) (ID:48325330)	Assessment of public hospital governance in Romania: Lessons from 10 case studies.		1		1
7	Etiaba (2018) (ID:48336789)	Strategic purchasing for universal health coverage: Examining the purchaser-provider relationship within a social health insurance scheme in Nigeria.		1	1	
8	Feldhaus (2018) (ID:48325319)	Effects of mixed provider payment systems and aligned cost sharing practices on expenditure growth management, efficiency, and equity: A structured review of the literature.			1	1
9	Fritsche (2018) (ID:48325381)	Methods to improve quality performance at scale in lower- and middle-income countries.	1		1	
10	Girdwood (2019) (ID:48325353)	Primary healthcare delivery models for uninsured low-income earners during the transition to National Health Insurance: Perspectives of private South African providers.		1		1
11	González-Block (2016) (ID:48325314)	[Financial allocations in the system for social protection in health in Mexico: Challenges for strategic purchasing].				1
12	Gorji (2018) (ID:48325312)	The challenges of strategic purchasing of healthcare services in Iran Health Insurance Organization: A qualitative study.		1		1
13	Honda (2020) (ID:48325364)	Payment arrangements for private healthcare purchasing under publicly funded systems in low- and middle-income countries: Issues and implications.			1	1
14	Ibe (2017) (ID:48325325)	Do beneficiaries' views matter in healthcare purchasing decisions? Experiences from the Nigerian tax-funded health system and the formal sector social health insurance program of the National Health Insurance Scheme.	1			1
15	Intaranongpal (2012) (ID:48325360)	The provincial health office as performance manager: Change in the local healthcare system after Thailand's universal coverage reforms.		1		1
16	Kundu (2018) (ID:48325392)	Analysis of multi drug resistant tuberculosis (MDR-TB) financial protection policy: MDR-TB health insurance schemes, in Chhattisgarh state, India.	1			

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Item	Title	Original search area result	Presence in sub-searches (1 = yes, blank = no)				Geography
			Benefits specification	Contracting arrangements	Provider payment	Performance monitoring	
17	Liu (2018) (ID:48325302)	Able to purchase? Agency problems in China's social health insurance system and the pitfalls of third-party strategic purchasing.				1	China
18	Maurya (2017) (ID:48336746)	Horses for courses: Moving India toward universal health coverage through targeted policy design.		1			India
19	Mbau (2018) (ID:48325310)	A critical analysis of health care purchasing arrangements in Kenya: A case study of the county departments of health.				1	Kenya
20	Mbau (2020) (ID:48325373)	Examining purchasing reforms toward universal health coverage by the National Hospital Insurance Fund in Kenya.	1	1	1	1	Kenya
21	Munge (2018) (ID:48336743)	A critical analysis of purchasing arrangements in Kenya: The case of the National Hospital Insurance Fund.	1	1		1	Kenya
22	Munge (2019) (ID:48325341)	A critical analysis of purchasing arrangements in Kenya: The case of micro health insurance.	1	1	1	1	Kenya
23	Obadha (2020) (ID:48336833)	Preferences of healthcare providers for capitation payment in Kenya: A discrete choice experiment.			1		Kenya
24	Obermann (2018) (ID:48336799)	The role of national health insurance for achieving UHC in the Philippines: A mixed methods analysis.	1		1	1	Philippines
25	Ogbuabor (2018) (ID:48325306)	Scaling-up strategic purchasing: Analysis of health system governance imperatives for strategic purchasing in a free maternal and child healthcare program in Enugu State, Nigeria.	1	1	1	1	Nigeria
26	Onwujekwe (2019) (ID:48325342)	Exploring effectiveness of different health financing mechanisms in Nigeria: what needs to change and how can it happen?	1			1	Nigeria
27	Pacharanarumol (2018) (ID:48336787)	Strategic purchasing and health system efficiency: A comparison of two financing schemes in Thailand.	1		1	1	Thailand
28	Preker (2005) (ID:48325357)	The role of purchasing in hospital performance.		1		1	Multiple
29	Pu (2018) (ID:48336793)	Provider payment to primary care physicians in China: Background, challenges, and a reform framework.	1	1	1		China
30	Ranson (2006) (ID:48325405)	Helping members of a community-based health insurance scheme access quality inpatient care through development of a preferred provider system in rural Gujarat.	1		1		India
31	Rao (2018) (ID:48325316)	Contracting non-state providers for universal health coverage: Learnings from Africa, Asia, and Eastern Europe.		1		1	Multiple
32	Ridde (2018) (ID:48325391)	Performance-based financing in Africa: Time to test measures for equity.	1		1		WHO Africa Region
33	Sanderson (2019) (ID:48325336)	What's needed to develop strategic purchasing in healthcare? Policy lessons from a realist review.				1	Multiple
34	Sapkota (2017) (ID:48325317)	Governance and purchasing function under Social Health Insurance in Nepal: Looking back and moving forward.				1	Nepal
35	Tangcharoensathien (2015) (ID:48325337)	Achieving universal health coverage goals in Thailand: The vital role of strategic purchasing.	1	1	1	1	Thailand

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Item	Title	Presence in sub-searches (1 = yes, blank = no)				
		Original search area result	Benefits specification	Contracting arrangements	Provider payment	Performance monitoring
36	Tangcharoensathien (2018) (ID:48336784)	Health systems development in Thailand: A solid platform for successful implementation of universal health coverage.			1	
37	Witter (2019) (ID:48325321)	(How) does RBF strengthen strategic purchasing of health care? Comparing the experience of Uganda, Zimbabwe and the Democratic Republic of the Congo.	1	1	1	1
38	Xu (2013) (ID:48325361)	Consumer choice among Mutual Healthcare Purchasers: A feasible option for China?	1			1
39	Yaghoubian (2020) (ID:48325421)	The requirements of strategic purchasing of health services for cancer patients: A qualitative study in Iran.	1		1	
40	Yip (2009) (ID:48325347)	Purchasing health care in China: Experiences, opportunities and challenges.		1		1
41	Yip (2019) (ID:48325300)	10 years of health-care reform in China: Progress and gaps in Universal Health Coverage.			1	1
42	Zizien (2019) (ID:48325338)	Contribution of the results-based financing strategy to improving maternal and child health indicators in Burkina Faso.			1	1
Total number of papers describing each purchasing function			18	17	22	29

Thailand

DRC, Uganda,
Zimbabwe

China

Iran

China

China

Burkina Faso

Appendix B. Summary of Findings

Country (Scheme)	Purchasing functions			Intermediate results <i>Efficiency (allocative and technical)</i>	Health system outcomes		
	<i>Benefits specification</i>	<i>Contracting</i>	<i>Provider payment</i> <i>Performance monitoring</i>		<i>Equity</i>	<i>Access</i>	<i>Quality</i> <i>Financial protection</i>
Argentina (Programa Sumar)		Contracts define services, payment, and performance metrics to be monitored. Providers have autonomy in financial decision making, giving them flexibility to respond to incentives. ⁴⁴	Resources are allocated to provinces using capitation payment adjusted by health and performance indicators, and providers are paid using fee-for-service.				Increased funding and autonomy at the provider level have increased investments in facilities to enable better service delivery. ¹
Cambodia (health equity funds, or HEFs)			HEFs compensate providers based on official performance-based financing flat-rate fees. ⁴⁵		HEFs have enabled more people to use public facilities and have increased access to health care by poor people by reducing financial barriers. ⁴⁶	HEFs have improved quality of care and increased accountability. ²	Improvements have been seen in access, quality of care, and accountability. ² HEFs have enabled more people to use public facilities and have increased access to health care by poor people by reducing financial barriers. ³

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Country (Scheme)	Purchasing functions			Intermediate results			Health system outcomes		
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality	Financial protection
China (social health insurance, or SHI)	The SHI system's contracting arrangements have resulted in substantial improvements in access to care and financial risk protection, with lower socioeconomic groups benefiting the most. ⁴⁷	Primary care providers receive a combination of input-based and performance-based payment. The performance-based managerial compensation framework pays hospital directors based on hospital performance against set targets. ⁴			The SHI system has allocated more resources to PHC and strengthened the PHC system, including for health education and promotion. This has helped address the increasing burden of noncommunicable diseases and disabilities associated with an aging population. ⁵ Redesigned price schedules for drugs and physician services and negotiation over drug prices between insurance plans and pharmaceutical companies have reduced costs. ⁵	Lower socioeconomic groups have benefited the most from the SHI system. ⁴	Access to health care has substantially improved. ⁴ Between 2010 and 2016, the hospital admission rate increased from 7.4% to 13.5% and the probability of seeing a doctor in the last 15 days increased from 16.2% to 22.7%. ⁴ Access and quality of care in urban and rural areas have improved. ⁴⁸	A combination of input-based and performance-based payment to primary care physicians has led to improved access and quality of care in urban and rural areas. ⁵	Financial protection has improved, with lower socioeconomic groups benefiting the most. ⁴

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Country (Scheme)	Purchasing functions			Intermediate results		Health system outcomes			
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality	Financial protection
Estonia (Estonia Health Insurance Fund, or EHIF)	The Health Insurance Act established criteria and a well-defined process for changing the benefit package. ²⁹	Contracting processes are well defined. Contracts are developed for 5-year cycles and include agreements on service quality and access, with detailed cost- and volume-based financial appendices. ⁶	The scheme uses performance-based payments layered on capitation.	97% of family physicians participate in the quality bonus system, which tracks disease prevention, chronic disease management, and prescribing patterns to ensure adherence to clinical guidelines. ⁶	Unnecessary tests and procedures reduce payments to providers. ⁶	In 2017, 94.1% of the population was covered by mandatory health insurance offered by the EHIF. Children and pensioners are exempt from contributions. ⁶ Disparities in membership exist across age groups and by gender. ⁶	Unmet need for a family doctor among people ages 16 and older is low, at 2%; the rate is similar in other income groups. Unmet need for specialists is higher, at 14%, and varies significantly across income groups; low-income groups have 20% unmet need and the highest quintile has 13% unmet need. ⁶	A quality bonus system has strengthened disease management for patients with hypertension, diabetes, and acute myocardial infarction. ⁶ In 2016, out-of-pocket payments accounted for 22.7% of total health expenditure; 50% to 60% of that was for pharmaceuticals. ⁶	No user fees are charged for primary care (except for home visits), to avoid financial barriers to accessing a family doctor or nurse. Co-payments are charged for specialist care. ⁶

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Country (Scheme)	Purchasing functions			Intermediate results		Health system outcomes		
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality
Ghana (National Health Insurance Scheme, or NHIS)	The NHIS benefit package is comprehensive.⁴⁹ Coverage of malaria services and medicines has improved utilization rates, with a 72% increase in the likelihood of receiving malaria medication.⁵⁰ Service delivery standards are specified. All NHIS members are entitled to the same benefit package.^{1,8-52} No systematic process exists for updating the benefit package.	Service delivery standards have led to improvements in quality, especially for malaria services.⁵³ Contracts include quality benchmarks. The National Health Insurance Authority includes quality benchmarks in provider contracts, such as average length of stay and minimum readmission period for inpatient services. By specifying these requirements in the contract and linking payments to achieving the benchmarks, Ghana gives providers a direct financial incentive to follow prescribed guidelines.			NHIS audits showed more than 1 million USD in savings from 2010 to 2015 from denial of claims for nonadherence to clinical guidelines or other treatment standards.⁷	The NHIS offers the same comprehensive benefit package (which includes about 95% of health services) to all members.	The NHIS benefit package has improved access to primary care.^{7,54} Coverage of malaria services and medicines has improved utilization rates, with a 72% increase in the likelihood of receiving malaria medication.⁷	Service delivery standards have led to improvements in quality, especially for malaria services.^{7,11} Payment delays reduce provider capacity to provide quality services.
							The NHIS offers the same comprehensive benefit package (which includes about 95% of health services) to all members, with few formal limits and no deductibles or copayments, to improve access and financial protection.^{1,8-10} Payment delays are one reason that providers charge informal payments, which pose a financial burden that reduces access.	

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Country (Scheme)	Purchasing functions			Intermediate results	Health system outcomes				
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality	Financial protection
Indonesia (Jaminan Kesehatan Nasional, or JKN)	The benefit package specifies service delivery standards to ensure quality of care. ^{8,55}					All JKN members are entitled to the same benefit package.		The benefit package specifies service delivery standards to ensure quality of care.	
Kenya (National Hospital Insurance Fund, or NHIF)	The evidence-based process for designing benefit packages includes explicit criteria that take into account the preferences and values of the population. Benefit packages differ by membership group. ^{8,12}				Service delivery improvements made by the Health Sector Services Fund (HSSF) were reversed when facility autonomy was withdrawn and budget execution decisions were recentralized.	NHIF benefit packages differ by membership group, which results in inequitable access to services. ²⁰ The packages are inadequately communicated, and benefits are unequally distributed across citizen groups. ^{15,56} Mixed provider payment streams reduced access for population groups whose purchaser has lower provider payment rates. ^{30,57}		HSSF increased resources to lower-level PHC facilities, leading to upgrades in infrastructure and equipment and overall improvement in service delivery and quality of care. ^{58,59}	

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Country (Scheme)	Purchasing functions			Intermediate results		Health system outcomes			
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality	Financial protection
Kyrgyzstan (Mandatory Health Insurance Fund, or MHIF)	The State-Guaranteed Benefit Package (SGBP) is a comprehensive package of health services that includes primary care, hospital care, palliative care, rehabilitative care, and most pharmaceuticals. ⁶⁰ Legislation provides broad criteria for the benefit package, and the Ministry of Health develops regulations for implementation of the package.	Contracting is mostly with public hospitals and some private facilities and private diagnostic centers. An accreditation framework is in place.	The scheme uses output-based payment (diagnosis-related group case-based payments). ⁶¹ Since 2018, the MHIF and providers have been able to carry forward unspent funds from all revenue sources. ¹⁶	The MHIF information system tracks service utilization and copayments. It uses the data to identify outliers for action, identify barriers to utilization, forecast income and expenditures, and estimate the impact of new policies (such as removing copayments for facility-based deliveries). ²¹	The share of health expenditures in the SGBP allocated to more cost-effective primary care increased from 29% in 2005 to 38% in 2009. This was the result of explicit budget allocation decisions at the MHIF, reflecting elements of strategic purchasing. MHIF (drugs, supplies, and food) increased from 20% in 2005 to 30% in 2009 as a result of optimizing health facility infrastructure; this allowed for a reduction in patient out-of-pocket expenditures. The average length of stay for hospital admissions was reduced by about 28% nationally during the same period. ²¹	All members have access to the same benefits. ²¹ Initial reductions in out-of-pocket spending after the introduction of the MHIF were particularly large among vulnerable, high-priority groups. ⁶²			From 2001 to 2006, the financial burden on the poorest 40% of the population declined significantly, the incidence of catastrophic payments was reduced, and geographic and financial barriers to access declined. In 2000, 11% of those needing care did not seek it due to financial reasons or distance; that fell to 3.1% in 2006, ⁶³ when the MHIF was introduced. However, it increased between 2006 and 2013. The proportion of patients who had to pay for drugs remained at about 51% to 53% after 2006. ⁶⁴ Between 2001 and 2006, informal payments for medicines and supplies declined significantly. ²¹

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Country (Scheme)	Purchasing functions			Intermediate results		Health system outcomes		
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality
Mexico (Seguro Popular)	Seguro Popular, which was discontinued in 2019, had an explicit package of services that guaranteed funding for priority interventions. ⁶⁵ The benefit package was designed through a consultative process with well-defined criteria for inclusion of interventions. ²⁵ No systematic process was in place for determining which new procedures, medicines, and devices to include in the benefit package. ^{66–68}	Performance agreements signed by state Seguro Popular managers and public providers focused on general targets that were not tied to specific funding or to sanctions for noncompliance. ²⁵	52% of resources were allocated through historical budgets tied to collective performance agreements. Public providers did not have autonomy to make decisions on how best to use resources; resources were allocated based on historical expenditure. ²⁵		Seguro Popular targeted the uninsured and subsidized the poorest income quintiles (I–IV). ²⁵ All members had the same benefits, but these were less comprehensive than in other insurance schemes in Mexico.	Seguro Popular reduced out-of-pocket expenses. ⁶⁹ But out-of-pocket payments remained the main source of health financing and accounted for 41.3% of total health expenditure in 2015. ²⁵		
Moldova (Mandatory Health Insurance, or MHI)	The comprehensive benefit package includes primary and secondary care services, emergency care, and dental services. ⁷⁰ Medicines are not covered, and informal charges have been reported for hospital services. ²⁹ MHI has a defined benefit package, while PHC services are provided for free through government budget financing.		The system uses prospective capitation plus retrospective quality incentive bonuses. ⁷¹			Members are subject to informal and out-of-pocket payments for uncovered medicines. In 2012, out-of-pocket payments accounted for 45% of total health expenditure. ⁷²		

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Country (Scheme)	Purchasing functions			Intermediate results			Health system outcomes		
	Benefits specification	Contracting	Provider payment	Performance monitoring	Efficiency (allocative and technical)	Equity	Access	Quality	Financial protection
Mongolia (Mandatory Health Insurance Scheme)	The mandatory health insurance scheme provides the same benefit package to all members. The scheme covers inpatient care; government budget funding covers outpatient care.		The system uses performance-based payment layered on capitation.	Performance-based payments are linked to health service targets.					Patients have high out-of-pocket costs for outpatient care. In 2012, these accounted for 66% of total out-of-pocket expenditures, of which two-thirds was spent on drugs. ⁷³
Nigeria (National Health Insurance Scheme)				Financial and human capacity constraints have hampered effective monitoring and made it difficult to ensure quality of care. ⁷⁴				Financial and human capacity constraints have hampered effective monitoring and made it difficult to ensure quality of care. ³⁵ Payment delays reduce provider capacity to deliver quality services. ¹³	Payment delays are one reason that providers charge informal payments, which reduce access to services and provider capacity to deliver quality services.
Philippines (PhilHealth)	Benefit packages differ between membership groups. PhilHealth lacks a systematic process for updating the benefit package, and additional benefits have been added incrementally, on an ad hoc basis, and as a result of lobbying. ⁷⁵								

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Country (Scheme)	Purchasing functions			Intermediate results <i>Efficiency (allocative and technical)</i>	Health system outcomes				
	<i>Benefits specification</i>	<i>Contracting</i>	<i>Provider payment</i>		<i>Performance monitoring</i>	<i>Equity</i>	<i>Access</i>	<i>Quality</i>	<i>Financial protection</i>
Thailand (UC Scheme)	The comprehensive benefit package has led to increased utilization of outpatient and inpatient services. Health technology assessments ensure that benefit package reviews are informed by evidence. All UC Scheme members are entitled to the same benefit package. No systematic process exists for determining which new procedures, medicines, and devices should be added to the benefit package.	The National Health Security Office contracts with the district health system's network of service providers to offer covered services. UC Scheme members are entitled to free services in the network in which they are registered.	Closed ended provider payment has contributed to lower expenditure than the Civil Servants Medical Benefits Scheme (CSMBS). ²⁸	Provincial health offices have implemented a performance management and quality assurance system to improve staff performance and quality outcomes. Monetary rewards for quality improvements are in place, and key performance indicators are linked to e-claims as the basis for reimbursement. ⁷⁷			The comprehensive benefit package has led to increased utilization of outpatient and inpatient services, ³ as well as greater probability that an ill person will receive formal treatment in the form of outpatient or inpatient services from a public provider. ⁷⁶ Significant improvements have been seen in access to care and outcomes for chronic noncommunicable diseases through continual care, medication, and home visits. ³ Linking providers to a specified population in terms of geographical proximity or range of services, coupled with a comprehensive benefit package, increased health service utilization and makes service delivery more equitable.		The comprehensive benefit package has led to reduced household health expenditure among poor households.

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Country (Scheme)	Purchasing functions			Intermediate results		Health system outcomes		
	<i>Benefits specification</i>	<i>Contracting</i>	<i>Provider payment</i>	<i>Performance monitoring</i>	<i>Efficiency (allocative and technical)</i>	<i>Equity</i>	<i>Access</i>	<i>Quality</i>
Vietnam (social health insurance)	Legislation established a comprehensive list of covered inpatient and outpatient services, including medicines, as well as co-payment rates. There is no transparent process for reviewing the benefit package or explicit criteria for adding benefits. ⁷⁸		Fee-for-service is the predominant payment method. Global budget, capitation, and diagnosis-related group payment are also used. ²⁷					In 2010, 60% of out-of-pocket spending was for pharmaceuticals. ⁴⁶